

CSSP 2026

FINAL CAPSTONE PROJECT

Website Security & Network Assessment

STUDENT	DETAILS
Name	Dinesh S
Batch	Batch 2
Submission Date	29/08/2026
Target	www.meesho.com

Academic assessment based on the evidence collected during the session. No exploitation, credential bypass, or destructive testing was performed.

1. Executive Summary

Assessment scope, target, evidence and reporting basis

This report documents a controlled website and network assessment of Meesho (www.meesho.com) using the evidence collected during the academic capstone exercise. The work covered domain information, DNS/IP resolution, connectivity, basic TCP port identification, HTTPS/TLS certificate review, response-header analysis, visible authentication flow, contact information, and security best-practice observations.

Assessment boundary: The report records observed facts only. Where a control could not be established from the available evidence, it is described as “not observed” rather than being labelled vulnerable or absent.

Area	Result
Domain / IP	meesho.com → 34.49.140.161
Connectivity	0% packet loss; latest observed average 43.539 ms
Open TCP ports	80/tcp HTTP; 443/tcp HTTPS
TLS	Valid and trusted wildcard certificate for *.meesho.com
HTTP technologies	HTTP/2; HTTP/3 advertised via Alt-Svc; Envoy indicator
Security headers	X-Content-Type-Options, X-Frame-Options, X-XSS-Protection, COOP, HSTS observed
Authentication	Phone-number login interface; SMS OTP reported

Authorization status: The supplied capstone template asks for authorization confirmation. No independent authorization document was supplied with the evidence in this report; therefore this report does not claim authorization beyond the student's stated academic exercise.

2. Task 1 — Domain Information

WHOIS, DNS and connectivity observations

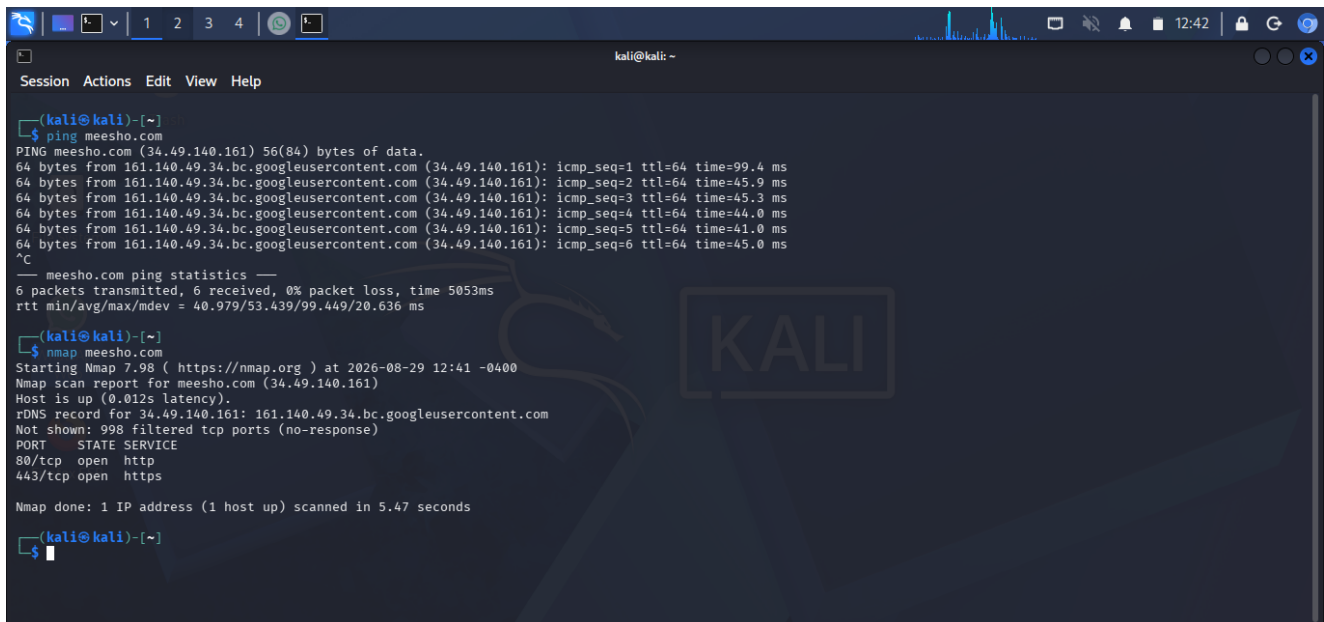
Field	Finding
Domain Name	meesho.com
Registrar	GoDaddy.com, LLC
Creation Date	2015-10-17
Registry Expiry	2028-10-17
Name Servers	NS-CLOUD-E1 through NS-CLOUD-E4.GOOGLEDOMAINS.COM
DNS A Record	34.49.140.161
Reverse DNS observed	161.140.49.34.bc.googleusercontent.com
DNSSEC	Unsigned

Connectivity: The collected ping evidence showed successful ICMP replies with 0% packet loss. The later six-packet sample displayed min/avg/max of 40.979 / 43.539 / 99.449 ms.

Commands used:

```
whois meesho.com
nslookup meesho.com
ping meesho.com
```

Evidence — Ping and Nmap terminal capture



```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali:~$ ping meesho.com  
PING meesho.com (34.49.140.161) 56(84) bytes of data:  
64 bytes from 161.140.49.34.bc.googleusercontent.com (34.49.140.161): icmp_seq=1 ttl=64 time=99.4 ms  
64 bytes from 161.140.49.34.bc.googleusercontent.com (34.49.140.161): icmp_seq=2 ttl=64 time=45.9 ms  
64 bytes from 161.140.49.34.bc.googleusercontent.com (34.49.140.161): icmp_seq=3 ttl=64 time=45.3 ms  
64 bytes from 161.140.49.34.bc.googleusercontent.com (34.49.140.161): icmp_seq=4 ttl=64 time=44.0 ms  
64 bytes from 161.140.49.34.bc.googleusercontent.com (34.49.140.161): icmp_seq=5 ttl=64 time=41.0 ms  
64 bytes from 161.140.49.34.bc.googleusercontent.com (34.49.140.161): icmp_seq=6 ttl=64 time=45.0 ms  
^C  
--- meesho.com ping statistics ---  
6 packets transmitted, 6 received, 0% packet loss, time 5053ms  
rtt min/avg/max/mdev = 40.979/53.439/99.449/20.636 ms  
kali@kali:~$ nmap meesho.com  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-29 12:41 -0400  
Nmap scan report for meesho.com (34.49.140.161)  
Host is up (0.012s latency).  
rDNS record for 34.49.140.161: 161.140.49.34.bc.googleusercontent.com  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
Nmap done: 1 IP address (1 host up) scanned in 5.47 seconds  
kali@kali:~$
```

3. Task 2 — Network Analysis

Basic network reachability and authorized-scope port identification

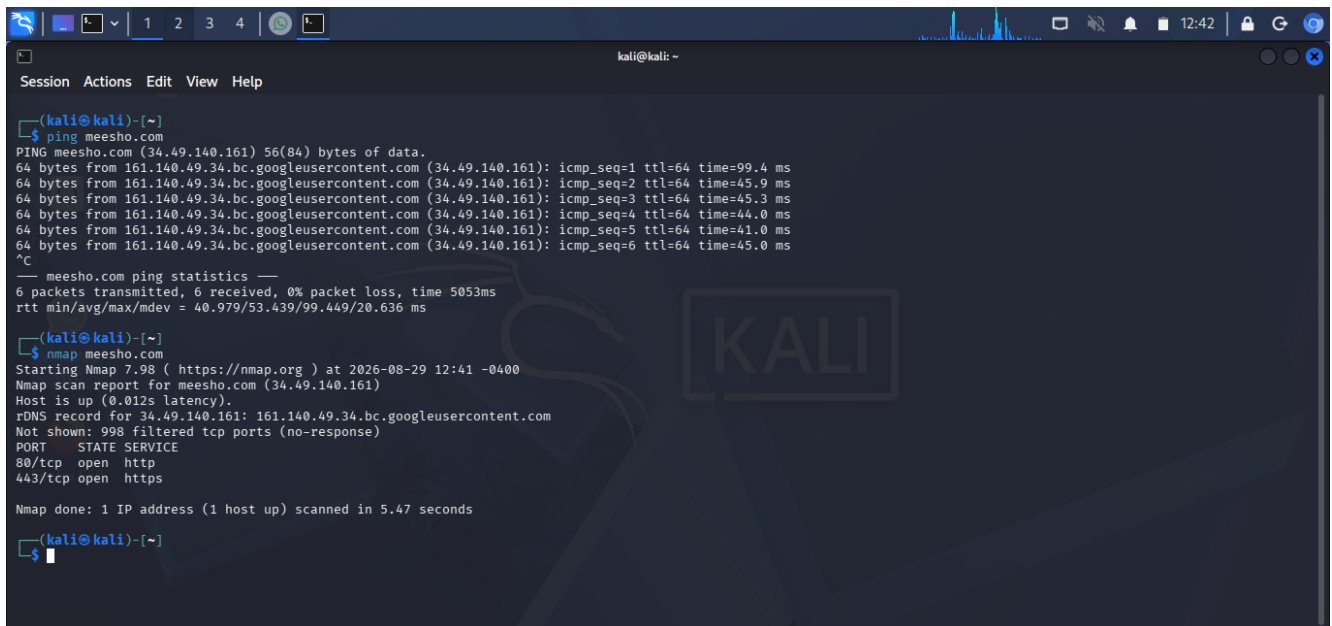
Metric	Observed result
Resolved IPv4	34.49.140.161
Host status	Up
Latest ping sample	6 transmitted / 6 received / 0% packet loss
Latest average RTT	43.539 ms
TCP 80	Open — HTTP
TCP 443	Open — HTTPS
Other default TCP ports	998 filtered / no response

Nmap evidence: The scan identified TCP ports 80 and 443 as open. The reverse-DNS name shown by Nmap was 161.140.49.34.bc.googleusercontent.com.

Commands used:

```
ping meesho.com
nmap meesho.com
```

Evidence — Nmap result



```
(kali@kali)~$ ping meesho.com
PING meesho.com (34.49.140.161) 56(84) bytes of data:
64 bytes from 161.140.49.34.bc.googleusercontent.com (34.49.140.161): icmp_seq=1 ttl=64 time=99.4 ms
64 bytes from 161.140.49.34.bc.googleusercontent.com (34.49.140.161): icmp_seq=2 ttl=64 time=45.9 ms
64 bytes from 161.140.49.34.bc.googleusercontent.com (34.49.140.161): icmp_seq=3 ttl=64 time=45.3 ms
64 bytes from 161.140.49.34.bc.googleusercontent.com (34.49.140.161): icmp_seq=4 ttl=64 time=44.0 ms
64 bytes from 161.140.49.34.bc.googleusercontent.com (34.49.140.161): icmp_seq=5 ttl=64 time=41.0 ms
64 bytes from 161.140.49.34.bc.googleusercontent.com (34.49.140.161): icmp_seq=6 ttl=64 time=45.0 ms
^C
--- meesho.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5053ms
rtt min/avg/max/mdev = 40.979/53.439/99.449/20.636 ms

(kali@kali)~$ nmap meesho.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-29 12:41 -0400
Nmap scan report for meesho.com (34.49.140.161)
Host is up (0.012s latency).
rDNS record for 34.49.140.161: 161.140.49.34.bc.googleusercontent.com
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 5.47 seconds

(kali@kali)~$
```

4. Task 3 — Website Analysis

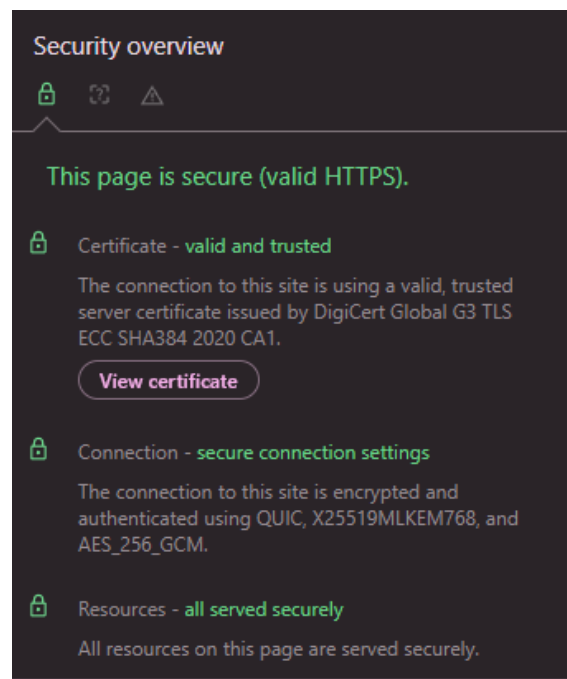
HTTPS, certificate and transport security

HTTP / HTTPS: HTTPS is enabled and the browser reported the page as secure with a valid HTTPS connection. The certificate viewer showed a wildcard certificate for *.meesho.com.

SSL Certificate:

Certificate field	Observed value
Common Name	*.meesho.com
Organization	Fashnear Technologies Private Limited
Issuer	DigiCert Global G3 TLS ECC SHA384 2020 CA1
Issuer Organization	DigiCert Inc
Issued On	18 February 2026, 5:30:00 AM
Expires On	20 February 2027, 5:29:59 AM
Status	Valid and trusted

Evidence — Browser security overview



Evidence — Certificate Viewer

Certificate Viewer: *.meesho.com

General

Details

Issued To

Common Name (CN)	*.meesho.com
Organization (O)	Fashnear Technologies Private Limited
Organizational Unit (OU)	<Not Part Of Certificate>

Issued By

Common Name (CN)	DigiCert Global G3 TLS ECC SHA384 2020 CA1
Organization (O)	DigiCert Inc
Organizational Unit (OU)	<Not Part Of Certificate>

Validity Period

Issued On	Wednesday, February 18, 2026 at 5:30:00 AM
Expires On	Saturday, February 20, 2027 at 5:29:59 AM

SHA-256 Fingerprints

Certificate	2171c22d0b237615c29b7bf9a06a1a18cec48f1244e637236d84700c281a6e05
Public Key	17dcffac1413d62f1759f7f08c85b0f2c4ef949fb8ad1e96e1cc5a3009f3d3ee

5. Task 3 — Technologies & Security Headers

Evidence from HTTP response headers

Technologies Identified

Technology / feature	Evidence / interpretation
HTTP/2	Response began with HTTP/2 200
HTTP/3 / QUIC availability	alt-svc: h3=":443"; ma=93600
Envoy indicator	x-envoy-upstream-service-time observed
HTTPS/TLS	Site accessed over HTTPS with valid certificate
Application framework	Not conclusively identified from captured headers

Security Headers Present

Security header	Observed value	Status
X-Content-Type-Options	nosniff	Present
X-XSS-Protection	1; mode=block	Present
X-Frame-Options	DENY	Present
Cross-Origin-Opener-Policy	same-origin	Present
Strict-Transport-Security	max-age=86400; includeSubDomains	Present
Content-Security-Policy	Not observed in tested response	Not observed
Referrer-Policy	Not observed in tested response	Not observed
Permissions-Policy	Not observed in tested response	Not observed

The clean terminal capture was filtered to relevant headers and did not expose the long cookie values present in the unfiltered response.

Evidence — Clean HTTP security-header capture

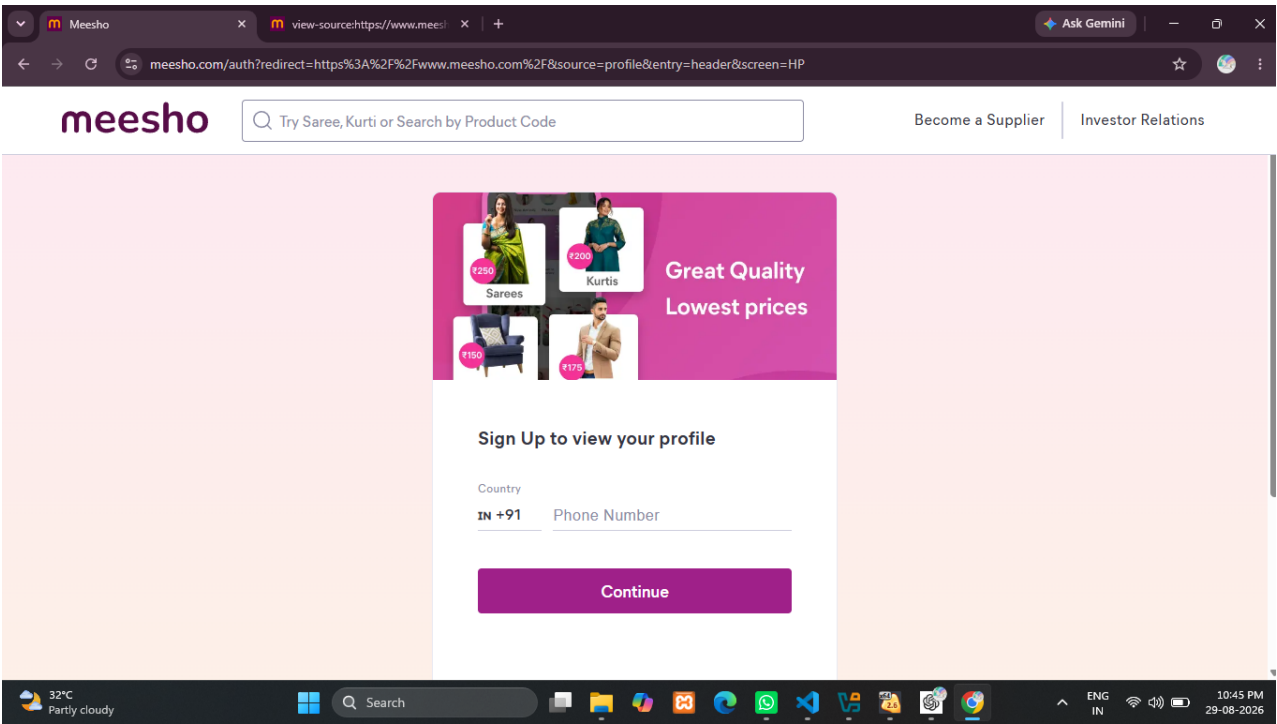
```
kali@kali: ~  
Session Actions Edit View WhatsApp Web Help  
set-cookie: bm.sz=E2799EEA6B4D4B9F8D35D1DB22C1CC5F-YAAQ15UvF5akPR2gAQAAmbWATgDepB+NoBGkAMKQr4EabWcj6WMPedCiuXXo2RHEhPW2P6tmSwcen+twN2I3W7kz0gujoOwrQU4Fh/9UZD8irzNT3Kr  
815aGChQiTnT/GlCj5nR3mJJkiCqsIarLONmSAkPMLDMCJaPnApS2PcPQU9lGIKDPfnOyggfIww2Qf4NYjH/D6vkTMqbH5Woak8bt12EqHir06RKLezdUQD8wLFe+31tgJ68MZ/KU8abicQhcJ+zJySkTsFor+fwT40boO  
lVettx3f2UFSy6dPX/ONAF6Vv2CV3BHCm7pOfaTCz90GkXILj6f1FPA68u12BuOjIZRrwhw0SMKjA+UsUwqGR3yYzJqSY7ovgmoh0PioEG3KasTxRv/v5g8fXbWX~3229232~3487557; Domain=.meesho.com; Path  
=/; Expires=Sat, 29 Aug 2026 21:10:52 GMT; Max-Age=14399  
  
(kali@kali)-[~]  
$ curl -sI https://www.meesho.com/ | grep -Ei '^(HTTP/|x-content-type-options:|x-xss-protection:|x-frame-options:|cross-origin-opener-policy:|strict-transport-securi  
ty:|content-security-policy:|referrer-policy:|permissions-policy:|alt-svc:|x-envoy-upstream-service-time:|)'  
HTTP/2 200  
x-content-type-options: nosniff  
x-xss-protection: 1; mode=block  
x-frame-options: DENY  
cross-origin-opener-policy: same-origin  
x-envoy-upstream-service-time: 21  
alt-svc: h3=":443"; ma=93600  
strict-transport-security: max-age=86400 ; includeSubDomains  
  
(kali@kali)-[~]  
$ curl -s https://www.meesho.com/ | grep -oiE '{0,80}(password policy|password requirements|password){0,120}' | head -20  
  
(kali@kali)-[~]  
$ curl -s https://www.meesho.com/ | grep -oiE '{0,80}(password policy|password requirements|password){0,120}' | head -20  
  
(kali@kali)-[~]  
$ curl -s https://www.meesho.com/ | grep -oi "password" | head  
  
(kali@kali)-[~]  
$ curl -sI https://www.meesho.com/ | grep -iE 'strict-transport-security|content-security-policy|referrer-policy|permissions-policy'  
strict-transport-security: max-age=86400 ; includeSubDomains  
  
(kali@kali)-[~]  
$
```

6. Task 4 — Security Observations

Authentication, password-policy observation and contact information

Required observation	Finding
Login Page Available?	Yes — profile access redirected to an authentication interface.
MFA Available?	SMS-based OTP was reported for login/sensitive actions. The evidence supports OTP use; it does
Password Policy Mentioned?	No password field or password requirements were observed in the accessible login interface.
Contact Information Exposed?	Yes — footer displayed Meesho Technologies Private Limited, Bangalore/Karnataka address and
Security Best Practices Observed?	HTTPS/TLS, HSTS, X-Content-Type-Options, X-Frame-Options, X-XSS-Protection and COOP were

Evidence — Login interface



7. Contact Information & Recommendations

Publicly visible business information and prioritized improvements

Contact information observed: The footer displayed Meesho Technologies Private Limited, an address in Bangalore, Karnataka, India, and the email address legal@meesho.com. It also provided legal/policy and social-media links.

Evidence — Website footer / contact information

Shop Non-Stop on Meesho

Trusted by crores of Indians
Cash on Delivery



Careers

Become a supplier

Hall of Fame

Sitemap

Legal and Policies

Meesho Tech Blog

Notices and Returns

Reach out to us



Contact Us

Meesho Technologies Private Limited
CIN: U62099KA2024PTC186568
3rd Floor, Wing-E, Helios Business
Park Kadubeesanahalli Village, Varthur Hobli,
Outer Ring Road Bellandur, Bangalore,
Bangalore South, Karnataka, India, 560103
E-mail address: legal@meesho.com
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Five Recommendations

1	Review and, where appropriate, implement a well-tuned Content Security Policy (CSP).
2	Consider adding Referrer-Policy and Permissions-Policy with settings appropriate to the application's requirements.
3	Review the HSTS max-age of 86400 seconds and consider whether a longer duration is appropriate for the production environment.
4	Continue strengthening OTP controls through expiry, retry limits, rate limiting and automated-abuse protection.
5	Maintain continuous monitoring of TLS, security headers, authentication controls and exposed network services.

8. Conclusion

Assessment outcome

The assessment successfully documented the required capstone areas for the selected target, Meesho. The evidence confirms DNS resolution to 34.49.140.161, successful connectivity, open HTTP/HTTPS services, a valid trusted TLS certificate, HTTP/2 and HTTP/3 availability indicators, and several browser security headers. The visible authentication flow uses a phone number, with SMS OTP reported for authentication. No password field or password requirements were observed in the accessible interface.

The assessment did not attempt exploitation, credential attacks, authentication bypass, or destructive activity. Findings such as “not observed” should be interpreted as limits of the captured evidence rather than proof that a control is absent across the entire production environment.

Overall assessment: Required capstone evidence collected and documented.

Appendix A — Commands Used

```
whois meesho.com
nslookup meesho.com
ping meesho.com
nmap meesho.com
curl -I https://www.meesho.com/
curl -sI https://www.meesho.com/ | grep -Ei 'relevant security headers'
curl -s https://www.meesho.com/ | grep -oi 'password' | head
```

Appendix B — Evidence Checklist

Evidence	Included
WHOIS / DNS / Ping	✓
Nmap port result	✓
HTTPS security overview	✓
TLS certificate viewer	✓
Clean security-header capture	✓
Login interface	✓
Contact/footer	✓